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Subject: Programming with JAVA	Sem: III AY: 2026-27
Assignment No: 17	
TITLE: Write a Java program to demonstrate the use of Swing and Applets	

Problem Statement

Write a Java program to demonstrate the use of Swing and Applets

Learning Objectives

To develop graphical user interfaces using Swing components and Applets.

Theory

Swing is Java's GUI toolkit that provides lightweight components such as buttons, labels, text fields, and menus. Applets were small Java programs embedded in web pages for interactive applications. Although Applets are obsolete, Swing remains an important framework for desktop application development. Swing supports platform-independent graphical interfaces.

Exercise

1. Create a simple Student Registration Form GUI using Swing.

```
import javax.swing.*;
```

```
public class StudentRegistration {  
    public static void main(String[] args) {
```

```
        JFrame frame = new JFrame("Student Registration Form");
```

```
        JLabel title = new JLabel("Student Registration Form");
```

```
JLabel nameLabel = new JLabel("Name:");
JLabel rollLabel = new JLabel("Roll No:");
JLabel courseLabel = new JLabel("Course:");
JLabel ageLabel = new JLabel("Age:");

JTextField nameField = new JTextField();
JTextField rollField = new JTextField();
JTextField courseField = new JTextField();
JTextField ageField = new JTextField();

JButton registerButton = new JButton("Register");

title.setBounds(120, 20, 200, 30);
nameLabel.setBounds(50, 70, 100, 30);
nameField.setBounds(150, 70, 200, 30);

rollLabel.setBounds(50, 110, 100, 30);
rollField.setBounds(150, 110, 200, 30);

courseLabel.setBounds(50, 150, 100, 30);
courseField.setBounds(150, 150, 200, 30);

ageLabel.setBounds(50, 190, 100, 30);
ageField.setBounds(150, 190, 200, 30);

registerButton.setBounds(150, 240, 100, 35);

frame.add(title);
frame.add(nameLabel);
frame.add(nameField);
```

```
frame.add(rollLabel);

frame.add(rollField);

frame.add(courseLabel);

frame.add(courseField);

frame.add(ageLabel);

frame.add(ageField);

frame.add(registerButton);


registerButton.addActionListener(e -> {

    JOptionPane.showMessageDialog(frame,

        "Registration Successful!\n" +

        "Name: " + nameField.getText() +

        "\nRoll No: " + rollField.getText() +

        "\nCourse: " + courseField.getText() +

        "\nAge: " + ageField.getText());

});


frame.setSize(420, 330);

frame.setLayout(null);

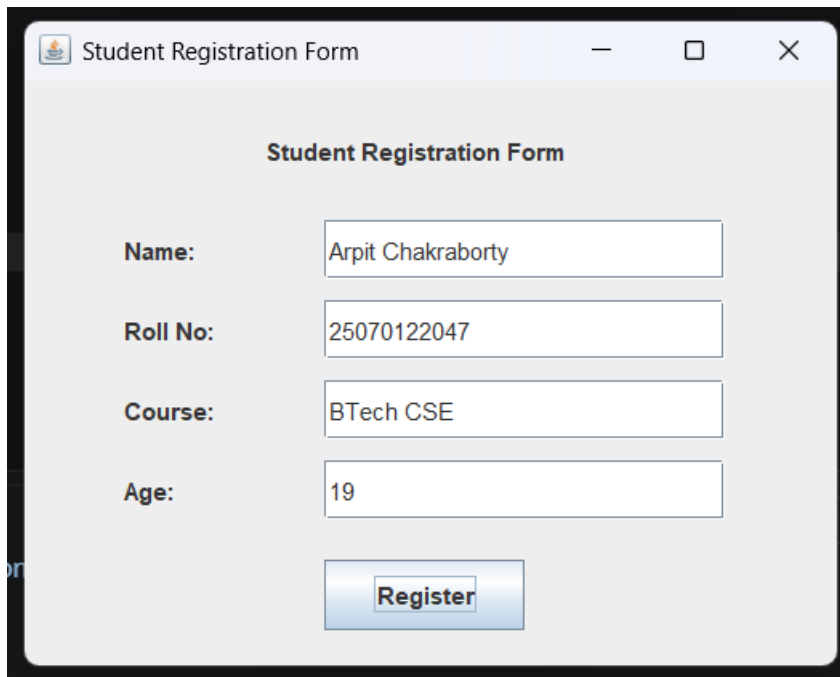
frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

frame.setVisible(true);

}

}
```

OUTPUT:



Student Registration Form

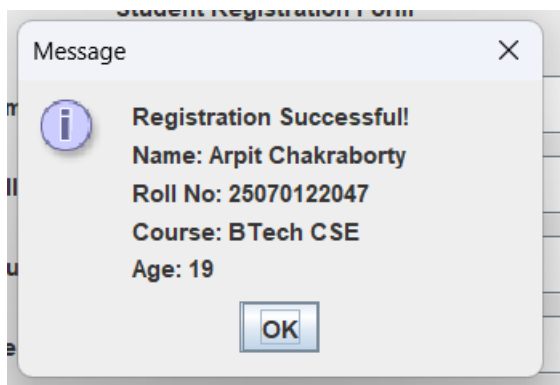
Name: Arpit Chakraborty

Roll No: 25070122047

Course: BTech CSE

Age: 19

Register



2. Create a simple Employee Registration Form using Java Swing GUI that takes employee details such as Employee ID, Name, Department, and Salary. Display the entered information in a dialog box

```
import javax.swing.*;
```

```
public class EmployeeRegistration {  
    public static void main(String[] args) {
```

```
        JFrame frame = new JFrame("Employee Registration Form");
```

```
        JLabel idLabel = new JLabel("Employee ID:");
```

```
JLabel nameLabel = new JLabel("Name:");  
JLabel deptLabel = new JLabel("Department:");  
JLabel salaryLabel = new JLabel("Salary:");
```

```
TextField idField = new TextField();  
TextField nameField = new TextField();  
TextField deptField = new TextField();  
TextField salaryField = new TextField();
```

```
Button submitButton = new Button("Submit");
```

```
idLabel.setBounds(40, 40, 120, 30);  
idField.setBounds(160, 40, 200, 30);
```

```
nameLabel.setBounds(40, 80, 120, 30);  
nameField.setBounds(160, 80, 200, 30);
```

```
deptLabel.setBounds(40, 120, 120, 30);  
deptField.setBounds(160, 120, 200, 30);
```

```
salaryLabel.setBounds(40, 160, 120, 30);  
salaryField.setBounds(160, 160, 200, 30);
```

```
submitButton.setBounds(160, 210, 100, 35);
```

```
frame.add(idLabel);  
frame.add(idField);  
frame.add(nameLabel);  
frame.add(nameField);  
frame.add(deptLabel);
```

```
frame.add(deptField);

frame.add(salaryLabel);

frame.add(salaryField);

frame.add(submitButton);


submitButton.addActionListener(e -> {

    JOptionPane.showMessageDialog(frame,

        "Employee Details\n\n" +

        "Employee ID: " + idField.getText() +

        "\nName: " + nameField.getText() +

        "\nDepartment: " + deptField.getText() +

        "\nSalary: " + salaryField.getText());

});


frame.setSize(420, 300);

frame.setLayout(null);

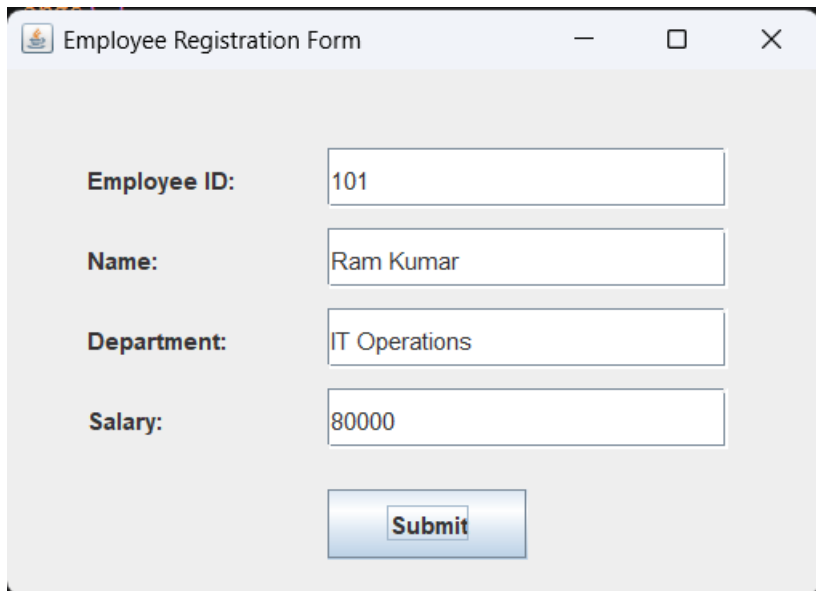
frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

frame.setVisible(true);

}

}
```

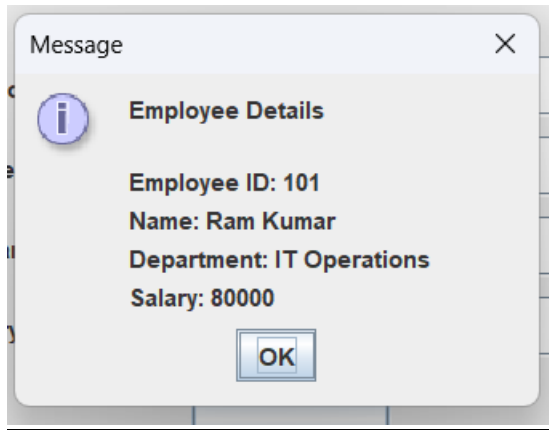
OUTPUT:



The screenshot shows a Java Swing window titled "Employee Registration Form". It contains four text input fields and a submit button. The fields are labeled "Employee ID:", "Name:", "Department:", and "Salary:". The values entered in the fields are "101", "Ram Kumar", "IT Operations", and "80000" respectively. The "Submit" button is located at the bottom of the form.

Label	Value
Employee ID:	101
Name:	Ram Kumar
Department:	IT Operations
Salary:	80000

Submit



Conclusion

We learned how to create graphical user interfaces using Java Swing. The Student and Employee Registration Forms successfully collect user details and display the entered information using dialog boxes.